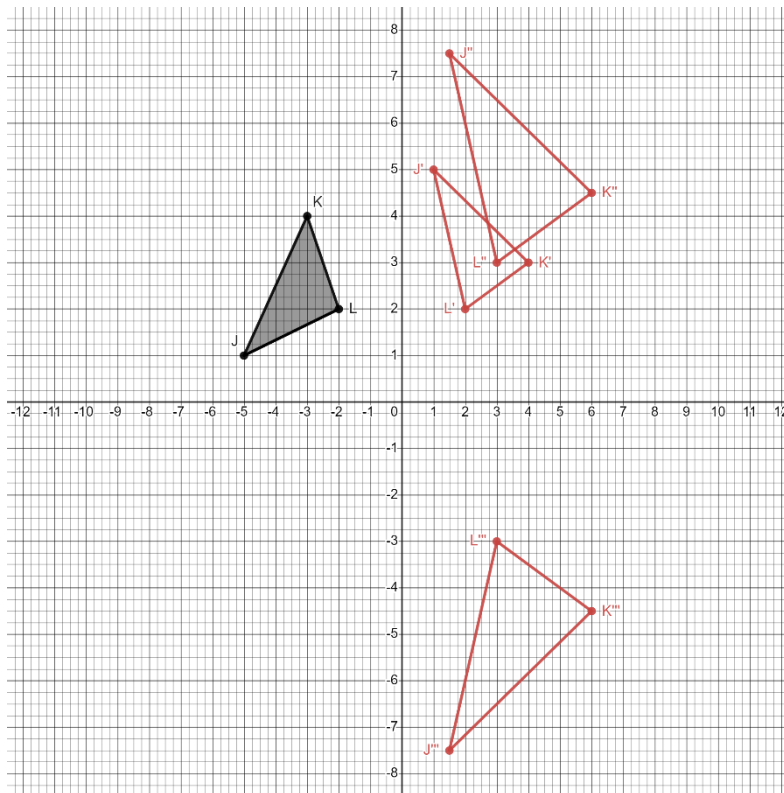


Geometry
Unit 6
Lesson H3 Practice

Name Answer Key
Date _____
Period _____

$\triangle JKL$ is shown.



Draw the result of each transformation in the following sequence:

- $\triangle J'K'L'$ as a result of $(x, y) \rightarrow (y, -x)$
- $\triangle J''K''L''$ as a result of $(x, y) \rightarrow \left(\frac{3}{2}x, \frac{3}{2}y\right)$
- $\triangle J'''K'''L'''$ as a result of $(x, y) \rightarrow (x, -y)$

Indicate the correct relationship for the resulting image:

- Image $\triangle J'K'L'$ is ☒ congruent
☐ similar to preimage $\triangle JKL$.
- Image $\triangle J''K''L''$ is ☐ congruent
☒ similar to preimage $\triangle JKL$.
- Image $\triangle J'''K'''L'''$ is ☐ congruent
☒ similar to preimage $\triangle JKL$.

$\triangle ABC$ and $\triangle DEF$ are shown.

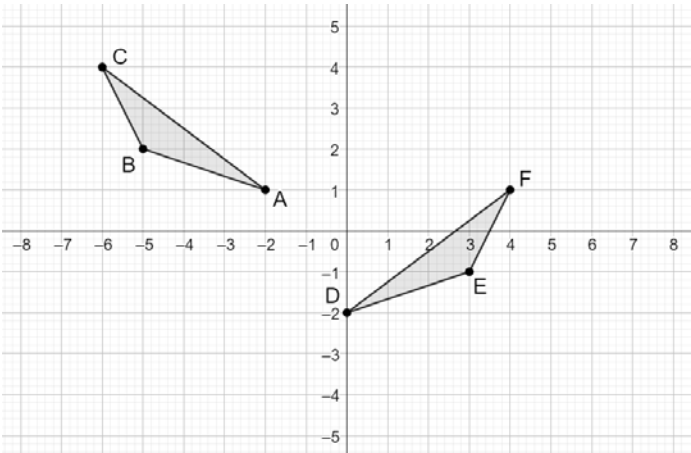
Assume: $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$

7. Write a series of transformations that will map $\triangle ABC$ onto $\triangle DEF$ so that $\triangle ABD \sim \triangle DEF$.

Answers may vary.

Example:

$(x,y) \rightarrow (-x,y)$ then $(x,y) \rightarrow (x-2,y-3)$



8. If it is given that $\triangle DEF$ is a result of $(x,y) \rightarrow (-x,y)$ then $(x,y) \rightarrow (x-2,y-3)$, then the triangles are

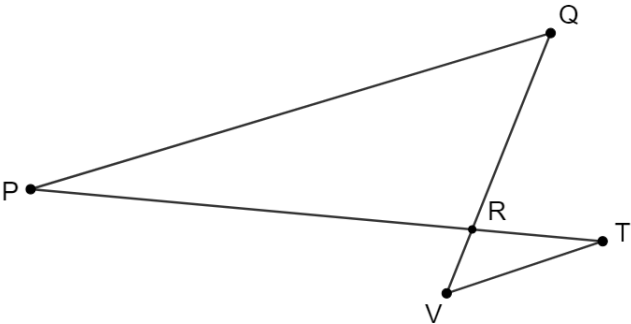
☒ congruent

☐ similar

.

$\triangle PQR$ and $\triangle TVR$ are shown.

Complete the table:



	If Given	True or False? $\triangle PQR \sim \triangle TVR$ by AA Similarity	Justification
9.	$\frac{TR}{PR} = \frac{VR}{QR}$	☐ True ☒ False	Triangles are similar by SAS similarity. Two sets of angles are not proven congruent.
10.	$\overline{PQ} \parallel \overline{VT}$	☒ True ☐ False	$\angle PQR \cong \angle TVR$ by alternate interior angles. $\angle QPR \cong \angle VTR$ by alternate interior angles. $\angle PRQ \cong \angle TRV$ by vertical angles. *Any combination of two of these three.